# Gravitational fields



9702 A Level Ch.17 | Chapter 17: Gravitational fields | Lesson 17.1 | Sections 17.1–17.7 | A Level only (not AS)

**Goal:** Use F = GMm/r², g = GM/r² and φ = −GM/r, and analyse circular orbits, including geostationary satellites.

force velocity

## 01 The big picture

A gravitational field is a region in which a mass feels a force. Its strength at a point is the force per unit mass on a small test mass placed there, $g = F/m$, in N kg$^{-1}$. The force is always attractive.

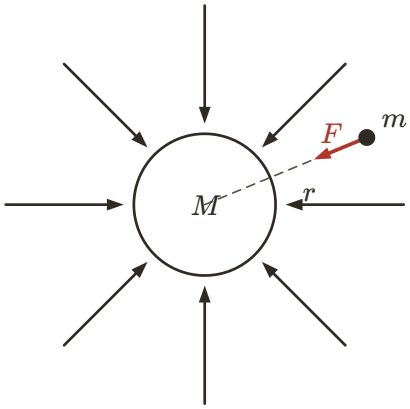
Two point masses attract with $F = GMm/r^2$: proportional to each mass and inversely proportional to the square of the distance between their centres. Outside a uniform sphere, the sphere acts as if all its mass were at its centre, so $r$ is always measured from the centre. Dividing by $m$ gives the field of a point mass, $g = GM/r^2$.

Field lines show the direction of the force on a mass; their spacing shows the strength. Round a planet they are radial and point inwards. Near the surface they are almost parallel and equally spaced: the field is nearly uniform, so $g \approx 9.81$ N kg$^{-1}$ and $\Delta E_\mathrm{p} = mg\Delta h$ work only for heights much smaller than the radius.

Gravitational potential $\varphi = -GM/r$ is zero at infinity and negative everywhere else: work must be done to pull a mass away. The potential energy of a mass $m$ is $E_\mathrm{p} = m\varphi = -GMm/r$, and moving it changes its energy by $m\Delta\varphi$.

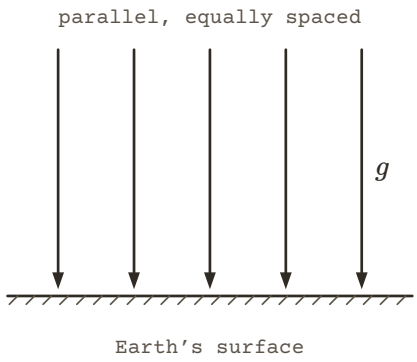
For an orbit, gravity provides the centripetal force. So $GMm/r^2 = mv^2/r$, giving $v = \sqrt{GM/r}$ and $T^2 \propto r^3$, whatever the satellite's mass.

**Fig. 1 · Radial field of a planet**




Field lines point inwards, towards the centre. They spread out with distance: the field gets weaker, as $1/r^2$.

**Fig. 2 · Near the surface: uniform**




Over heights much smaller than the radius, the field lines are parallel and equally spaced, so $g$ is constant.

**Gravitational field strength** [paraphrase]
The gravitational force per unit mass on a small test mass at that point, $g = F/m$.

**Newton's law of gravitation** [paraphrase]
Two point masses attract each other with a force proportional to the product of their masses and inversely proportional to the square of their separation.

**Gravitational potential (syllabus)** [syllabus wording]
"The work done per unit mass in bringing a small test mass from infinity to the point."

**Geostationary orbit (syllabus)** [syllabus wording]
"A satellite in a geostationary orbit remains at the same point above the Earth's surface, with an orbital period of 24 hours, orbiting from west to east, directly above the Equator."

## 02 Deriving the equations

### D1 Field strength of a point mass

Put a small test mass $m$ at distance $r$ from the centre of a mass $M$, and use the definition $g = F/m$.

1. Newton's law of gravitation gives the force on the test mass:
$$F = \frac{GMm}{r^2}$$
2. Field strength is force per unit mass, so divide by $m$; the test mass cancels:
$$g = \frac{F}{m} = \frac{GM}{r^2}$$
3. At the surface, $r = R$ and $g_0 = GM/R^2$. Dividing the two gives how $g$ falls with distance (inverse square):
$$\frac{g}{g_0} = \frac{R^2}{r^2}$$

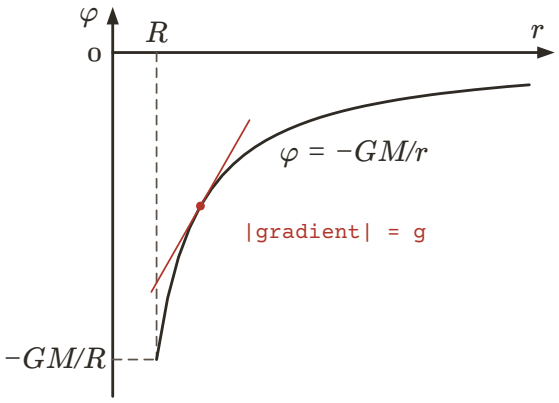
RESULT

$$g = \frac{GM}{r^2}$$

### D2 Potential and potential energy

Potential is zero at infinity. A mass coming in from infinity is pulled by the field, so the work done on it per unit mass is negative.

**Fig. 3 · Potential against distance**




φ rises from −GM/R at the surface towards zero at infinity. The gradient of the graph at any point is the field strength there.

1. The potential of a point mass:
$$\varphi = -\frac{GM}{r}$$
2. The potential energy of a mass $m$ at distance $r$ is $m$ times the potential:
$$E_\mathrm{p} = m\varphi = -\frac{GMm}{r}$$
3. Moving from $r_1$ out to $r_2$ raises the potential (towards zero), so it takes energy:
$$\Delta E_\mathrm{p} = m\Delta\varphi = GMm\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$$
4. The field strength is the size of the gradient of the potential–distance graph:
$$g = -\frac{\Delta\varphi}{\Delta r}$$

RESULT

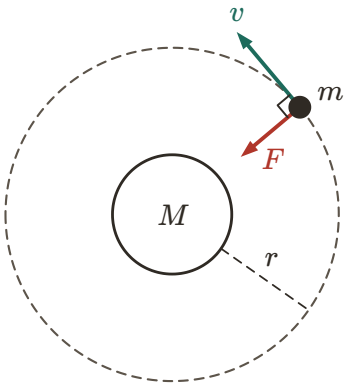
$$\varphi = -\frac{GM}{r} \qquad E_\mathrm{p} = -\frac{GMm}{r}$$

## 02 Deriving the equations (continued)

### D3 Circular orbits and T² ∝ r³

A satellite of mass $m$ orbits a planet of mass $M$ at radius $r$. Gravity is the only force on it, so gravity provides the centripetal force.

**Fig. 4 · A satellite in a circular orbit**




Gravity is the only force on the satellite: it points to the centre of the orbit and provides the centripetal force.

1. Gravitational force = centripetal force:
$$\frac{GMm}{r^2} = \frac{mv^2}{r}$$
2. Cancel $m$ and one $r$: the orbital speed does not depend on the satellite's mass.
$$v = \sqrt{\frac{GM}{r}}$$
3. Substitute $v = 2\pi r/T$ and square:
$$\frac{4\pi^2 r^2}{T^2} = \frac{GM}{r} \implies T^2 = \frac{4\pi^2}{GM}\,r^3$$
4. For satellites of the same planet $4\pi^2/GM$ is constant, so $T^2 \propto r^3$.

**RESULT**

$$v = \sqrt{\frac{GM}{r}} \qquad T^2 = \frac{4\pi^2 r^3}{GM}$$

### D4 Geostationary orbit

A geostationary satellite stays above the same point on the Equator: it orbits west to east with the Earth's rotation, with period $T = 24\ \text{h} = 86\,400\ \text{s}$.

1. Cross-multiply $T^2 = 4\pi^2 r^3/(GM)$ and divide by $4\pi^2$:
$$r^3 = \frac{GMT^2}{4\pi^2} = \frac{6.67\times10^{-11}\times 5.97\times10^{24}\times 86\,400^2}{4\pi^2} = 7.53\times10^{22}\ \text{m}^3$$
2. Take the cube root, then subtract the Earth's radius for the height:
$$r = 4.22\times10^{7}\ \text{m}, \qquad h = r - R = 3.59\times10^{7}\ \text{m}$$

**RESULT**

$$r \approx 4.2\times10^{7}\ \text{m} \quad (h \approx 3.6\times10^{7}\ \text{m})$$

## 03 Key equations

**Field strength (definition)**

$$g = \frac{F}{m}$$

**Newton's law of gravitation**

$$F = \frac{GMm}{r^2}$$

**Field of a point mass**

$$g = \frac{GM}{r^2}$$

r from the centre

**Gravitational potential**

$$\varphi = -\frac{GM}{r}$$

zero at infinity

**Potential energy**

$$E_\mathrm{p} = m\varphi = -\frac{GMm}{r}$$

**Change in potential energy**

$$\Delta E_\mathrm{p} = m\,\Delta\varphi$$

**Orbital speed**

$$v = \sqrt{\frac{GM}{r}}$$

from GMm/r² = mv²/r

**Orbital period (T² ∝ r³)**

$$T^2 = \frac{4\pi^2 r^3}{GM}$$

**Near the surface only**

$$\Delta E_\mathrm{p} = mg\,\Delta h$$

when Δh ≪ R

**Symbols and units**

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| $G$ | gravitational constant, 6.67 × 10⁻¹¹ | $\mathrm{N\,m^2\,kg^{-2}}$ | $M$ | mass of the planet or star | $\mathrm{kg}$ | $m$ | mass of the test mass or satellite | $\mathrm{kg}$ |
| $r$ | distance from the centre | $\mathrm{m}$ | $R$ | radius of the planet | $\mathrm{m}$ | $g$ | gravitational field strength | $\mathrm{N\,kg^{-1}}$ |
| $\varphi$ | gravitational potential | $\mathrm{J\,kg^{-1}}$ | $E_\mathrm{p}$ | gravitational potential energy | $\mathrm{J}$ | $v$ | orbital speed | $\mathrm{m\,s^{-1}}$ |
| $T$ | orbital period | $\mathrm{s}$ | $M_\mathrm{E}$ | Earth's mass, 5.97 × 10²⁴ | $\mathrm{kg}$ | $R_\mathrm{E}$ | Earth's radius, 6.37 × 10⁶ | $\mathrm{m}$ |

## 04 Sample problems with solutions

**Sample 1 · Field strength above the surface**

The International Space Station orbits 4.1 × 10⁵ m above the Earth's surface. Calculate the gravitational field strength there. Use the Earth data in the symbols list.

**Step 1.** $r$ is measured from the centre of the Earth, not from the surface:

$$r = R + h = 6.37 \times 10^6 + 4.1 \times 10^5 = 6.78 \times 10^6\ \mathrm{m}$$

**Step 2.** Substitute into $g = GM/r^2$:

$$g = \frac{6.67 \times 10^{-11} \times 5.97 \times 10^{24}}{(6.78 \times 10^6)^2} = 8.66\ \mathrm{N\,kg^{-1}}$$

**Step 3.** Sense check: a little less than 9.81 N kg⁻¹. Astronauts feel weightless because they are in free fall, not because $g$ is zero.

**Answer:** $g = 8.7\ \mathrm{N\,kg^{-1}}$

Unit check: $\frac{\mathrm{N\,m^2\,kg^{-2}\times kg}}{\mathrm{m^2}} = \mathrm{N\,kg^{-1}}$ ✓

## 04 Sample problems with solutions (continued)

**Sample 2 · Energy to lift a probe a long way up**

A 500 kg probe is lifted from the Earth's surface to a height equal to the Earth's radius. Calculate the increase in its gravitational potential energy.

**Step 1.** Potential at the surface ($r = R$) and at height $R$ ($r = 2R$):

$$\varphi_R = -\frac{GM}{R} = -6.25 \times 10^{7}\ \text{J kg}^{-1}, \qquad \varphi_{2R} = -3.13 \times 10^{7}\ \text{J kg}^{-1}$$

**Step 2.** Change in potential energy:

$$\Delta E_\text{p} = m\Delta\varphi = 500 \times 3.13 \times 10^{7} = 1.56 \times 10^{10}\ \text{J}$$

**Step 3.** Why not $mgh$? It gives $3.1 \times 10^{10}$ J, twice too much, because $g$ falls to a quarter of its surface value over this height.

**Answer:** $\Delta E_\text{p} = 1.6 \times 10^{10}\ \text{J}$

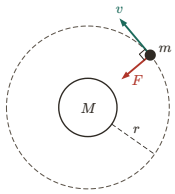
Unit check: $\text{kg} \times \text{J kg}^{-1} = \text{J}$ ✓

**Sample 3 · Speed and period of a satellite**

A satellite orbits 2.0 × 10⁶ m above the Earth's surface. Calculate its orbital speed and its period.

**Step 1.** Radius of the orbit, from the centre:

$$r = 6.37 \times 10^{6} + 2.0 \times 10^{6} = 8.37 \times 10^{6}\ \text{m}$$




see Fig. 4

**Step 2.** Gravity provides the centripetal force, so:

$$v = \sqrt{\frac{GM}{r}} = \sqrt{\frac{6.67 \times 10^{-11} \times 5.97 \times 10^{24}}{8.37 \times 10^{6}}} = 6.90 \times 10^{3}\ \text{m s}^{-1}$$

**Step 3.** One orbit is a distance $2\pi r$:

$$T = \frac{2\pi r}{v} = \frac{2\pi \times 8.37 \times 10^{6}}{6.90 \times 10^{3}} = 7.62 \times 10^{3}\ \text{s}$$

**Answer:** $v = 6.9 \times 10^{3}\ \text{m s}^{-1}$, $T = 7.6 \times 10^{3}$ s (about 2.1 hours)

Unit check: $\sqrt{\frac{\text{N m}^2\,\text{kg}^{-2} \times \text{kg}}{\text{m}}} = \sqrt{\text{m}^2\,\text{s}^{-2}} = \text{m s}^{-1}$ ✓

**Sample 4 · Weighing the Earth with the Moon**

The Moon orbits the Earth at a radius of 3.84 × 10⁸ m with a period of 27.3 days. Calculate the mass of the Earth.

**Step 1.** Period in seconds:

$$T = 27.3 \times 86\,400 = 2.36 \times 10^{6}\ \text{s}$$

**Step 2.** $M$ is in the denominator and $T^2$ is the lone term on the other side: swap them in one step (cross-multiplication).

$$T^2 = \frac{4\pi^2 r^3}{GM} \implies M = \frac{4\pi^2 r^3}{GT^2}$$

**Step 3.** Substitute:

$$M = \frac{4\pi^2 \times (3.84 \times 10^{8})^3}{6.67 \times 10^{-11} \times (2.36 \times 10^{6})^2} = 6.02 \times 10^{24}\ \text{kg}$$

**Answer:** $M = 6.0 \times 10^{24}$ kg (close to the accepted $5.97 \times 10^{24}$ kg)

Unit check: $\frac{\text{m}^3}{\text{N m}^2\,\text{kg}^{-2} \times \text{s}^2} = \frac{\text{m}^3\,\text{kg}^2}{\text{kg m}^3} = \text{kg}$ ✓

## 05 Practice problems

**P1** Calculate the gravitational force between two 70 kg people whose centres are 1.0 m apart.
Answer: $3.3 \times 10^{-7}$ N

**P2** The Moon has mass $7.35 \times 10^{22}$ kg and radius $1.74 \times 10^{6}$ m. Calculate the field strength at its surface.
Answer: 1.6 N kg$^{-1}$

**P3** A planet's field strength at distance r is g. What is it at distance 3r?
Answer: g/9

**P4** At what height above the Earth's surface is g half its surface value?
Answer: $2.6 \times 10^{6}$ m

**P5** Calculate the gravitational potential at the Earth's surface.
Answer: $-6.3 \times 10^{7}$ J kg$^{-1}$

**P6** The potential at a point is $-4.0 \times 10^{7}$ J kg$^{-1}$. Calculate the potential energy of a 200 kg satellite there.
Answer: $-8.0 \times 10^{9}$ J

**P7** Calculate the speed of a satellite in a circular orbit of radius $7.0 \times 10^{6}$ m round the Earth.
Answer: $7.5 \times 10^{3}$ m s$^{-1}$

**P8** Two satellites orbit the same planet; one orbit has 4 times the radius of the other. Calculate the ratio of their periods.
Answer: 8

**P9** Phobos orbits Mars at a radius of $9.38 \times 10^{6}$ m with a period of 7.65 h. Calculate the mass of Mars.
Answer: $6.4 \times 10^{23}$ kg

**P10** Calculate the height above the Earth's surface of a geostationary satellite.
Answer: $3.6 \times 10^{7}$ m

## 06 Other useful information

### Common mistakes

✗ **Wrong:** Use the height above the surface as r.
✓ **Right:** r is the distance from the centre: r = R + h.

✗ **Wrong:** Use mgh for any change in height.
✓ **Right:** Only when g is constant (h ≪ R). Otherwise use ΔEp = mΔφ.

✗ **Wrong:** Astronauts float because there is no gravity in orbit.
✓ **Right:** g is still about 8.7 N kg$^{-1}$ at the ISS; they are in free fall with the station.

✗ **Wrong:** Gravitational potential is positive, like height.
✓ **Right:** It is zero at infinity and negative everywhere else; it rises towards zero as you move away.

✗ **Wrong:** A heavier satellite must orbit faster.
✓ **Right:** m cancels: v = √(GM/r) depends only on M and r.

### Exam tips

- Write r = R + h at the start of every calculation above a planet.
- Orbits: start from $GMm/r^2 = mv^2/r$ (or $mr\omega^2$), then cancel m.
- Ratio questions: use $g \propto 1/r^2$, $\varphi \propto -1/r$ and $T^2 \propto r^3$; no constants needed.
- Convert periods to seconds: 24 h = 86 400 s.

### Explore

- Practice: field strength of a point mass
- Practice: gravitational potential
- Practice: orbits (32 questions)

### What the syllabus asks

- 9702 topic 13, Gravitational fields: field strength as force per unit mass, field lines, $F = Gm_1m_2/r^2$, $g = GM/r^2$, $\varphi = -GM/r$, $E_p = -GMm/r$, circular orbits and geostationary orbits. Coursebook Chapter 17.
- A Level only: not part of AS (AS uses $\Delta E_p = mg\Delta h$ in a uniform field).
- Outside a uniform sphere, treat its mass as a point mass at its centre.
- Gravitational and electric fields are compared in Chapter 22.